

CS4001 - Professional Practices in IT

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Course Instructor: Anaum Hamid

Email address: anaum.hamid@nu.edu.pk

Continuing Profession

- ▶ Profession vs Occupation
 - ▶ Commonly people use these two words interchangeably. Are they the words meaning the same? Are they synonyms?
 - ▶ What is the meaning of occupation?
 - ▶ A person involved in a job and earning money is having that job as an occupation

Example:

A doctor teaching in a school, college or university has an occupation of teaching. A carpenter, a driver etc..

Differences between Occupation and Profession

- ▶ A profession needs extensive training and specialized knowledge. On the other hand, an occupation does not need any extensive training.
- ▶ A profession can be called an occupation when a person is paid for his particular skills, and his deep knowledge. Persons engaged in an occupation are not paid for their knowledge, but only for what they produce.
- ▶ Unlike a person engaged in an occupation, a professional has to undergo higher education.
- ▶ A profession tends to be autonomous, whereas, for an occupation, no one has autonomous power; he or she is supervised by another person.
- ▶ Unlike occupation, a profession demands that the responsibility lies with the individual.
- ▶ A profession is guided through certain ethical codes, and regulated by certain statute.

Characteristics of Profession

Getting to the definition of profession ; there is none called hard and fast definition. Thus professions are often associated with some characteristics briefed as following:

- ▶ **Mastery of an Esoteric Body of knowledge:** usually acquired through higher education
 - ▶ A member of the profession needs this body of knowledge in order to practice
 - ▶ A related characteristic of professions is that they often embrace a division between researchers and practitioners

Characteristics.....

- ▶ **Autonomy:** Professionals generally have a great deal of autonomy, justified by having an esoteric knowledge
 - ▶ Professions typically have autonomy both at the collective level as well as in individual practice
 - ▶ Profession sets its own standards rather than taking orders or suggestion from outsiders
- ▶ **Formal Organization:** Generally there exists a professional organization that:
 - ▶ Controls admission to the profession and sets standards
 - ▶ Involved in licensing & expelling its individual members

Characteristics.....

► **Code of ethics**

- There is a code of ethics & members of professions must adhere to the code no matter what their employment context (take an oath)

And an important characteristic mostly taken for granted:-

► **Social function:** Professions generally fulfill an important social function and for the welfare of the society

- Medicine (promoting health)
- lawyers not alone but play an important role in justice

PROFESSIONAL BODIES

A profession is typically organized into one or more professional bodies. What is a professional body?

- ▶ A **professional association** (also called a **professional body**, **professional organization**, or **professional society**) is usually a nonprofit organization seeking to further a particular profession, the interests of individuals engaged in that profession and the public interest. OR
- ▶ A professional body is a group of people in a learned occupation who are entrusted with maintaining control or oversight of the practice of the occupation
- ▶ A professional body usually starts by a group of people coming together because of a shared interest in a particular type of activity.

Functions of Professional bodies

As the professional body matures, it is likely to develop a range of functions, of which the following are the most important:

- ▶ establishing a code of conduct to regulate the way members of the body behave in their professional lives and a disciplinary procedure to discipline members who breach(fail to follow or conduct) this code;
- ▶ establishing mechanisms for disseminating(spreading) knowledge of good practice and new developments to its members, typically through publications and conferences but increasingly also through the use of the worldwide web;
- ▶ setting standards of education and experience that must be met by people wishing to become members of the body;
- ▶ advising government and regulatory bodies about matters within its area of expertise.

RESERVATION OF TITLE AND FUNCTION

- ▶ The use of the name of the profession may be restricted to those people who are appropriately qualified. A restriction of this sort is called ***reservation of title***.
- ▶ The law may state that certain activities are restricted to people with appropriate qualifications. This is called ***reservation of function***.

Is Computing A Profession

- ▶ Where does computing fit in the process of professionalization?
- ▶ What does the field of computing look like when described in terms of the mentioned characteristics?
 - ▶ The field of computing is so diverse and complex
 - ▶ The range of those who are called computer professionals is extremely broad
 - ▶ Nothing in common except that, in one way or another, they involve the use of computer
 - ▶ At first glance, it seem that computing possesses all of these characteristics, though in complex ways

Is Computing A Profession

- ▶ **Are Computer Professionals “Professionals”?**
 - ▶ Computer professionals have mastered an esoteric body of knowledge
 - ▶ Computer professionals have varying degrees of autonomy depending on where they work
 - ▶ There is no single organization governing the profession of computing (ACM, IEEE)
 - ▶ Computing is an activity which supports social institutions which in turn are aimed at fulfilling a variety of social functions
 - ▶ Computing does not appear to be a 'profession' in the strictest sense, the way law and medicine are professions, but it is more of a 'profession' than many other occupations.

SOFTWARE ENGINEERING

- ▶ Software engineering appears to be one area of computing that is emerging as a distinct profession within the field of computing.
- ▶ This trend seems to have been created by the initiative of individuals in the field concerned about the quality and safety of the software being produced and sold.
- ▶ Developing software engineering as a distinct field involves several of the activities.
 - ▶ For example, it means identifying a unique body of knowledge that a person must possess to be competent software engineer.
 - ▶ It means developing educational requirements (curriculum) such that the person who meets the requirements is more likely to produce a quality, safe software than someone without training.

SOFTWARE ENGINEERING

- ▶ It means developing mechanisms for licensing of members.
- ▶ This will include identifying or creating the proper organization for issuing licenses and identifying requirements for obtaining a license, such as passing an exam or acquiring a certain number of years of experience.
- ▶ Another requirement for professionalization is a code of ethics.
- ▶ The state of Texas has boldly taken the first steps in this process and has established software engineering licensing in its state.
- ▶ The Texas initiative is a serious attempt at setting standards in the field.

SOFTWARE DEVELOPMENT AS ENGINEERING

- ▶ The development of software and information systems in general is now usually regarded as a branch of engineering and the people who practice it are considered to be engineers.
 - ▶ Traditional engineers design and build a wide variety of objects – dams, bridges, aeroplanes, cars, radio and television transmitters and receivers, computers, plants to make fertilizer or plastics, and so on.

Characteristic of engineering:

- ▶ There are two constraints that apply to all such activities and which can be regarded as characteristic of engineering:
 - ▶ First engineering involves designing and building things that must work properly, that is, must meet a set of predetermined requirements concerning their functionality, their performance, and their reliability;
 - ▶ the process of designing and building the object must be completed within specified constraints of time and budget.

THE STATUS OF ENGINEERS

The legal status of the engineering profession varies a lot from one country to another. however, the position is that:

- ▶ it is illegal to call yourself an engineer in a given state unless you are registered with the State Engineers Registration Board;
- ▶ it is illegal for a company to use the word 'engineering' in its name unless it employs at least one registered engineer;
- ▶ academic programmes including the term engineering in their title must be taught mostly by registered engineers;
- ▶ it is illegal to carry out engineering work except under the supervision of a registered engineer.